

Teaching Statement

Mohammad Ryiad Al-Eiadeh

Teaching Motivation

Throughout my academic journey, I have viewed teaching as one of the most meaningful parts of academia. Teaching is more than delivering content; it is an opportunity to inspire students, build confidence, and help them discover how knowledge can be transformed into solutions for real problems. My motivation comes from seeing students progress from uncertainty to independence and from helping them realize that they are capable of solving challenging technical problems through structured thinking and persistence.

My teaching philosophy is guided by two complementary goals: first, to develop analytical and scientific maturity through critical thinking, mathematical reasoning, and problem formulation; and second, to prepare students for real-world impact by connecting theory to practical engineering and computing applications. These principles have been shaped by my experiences as a Graduate Teaching Assistant at Purdue University.

Teaching Interests and Philosophy

I am excited to teach courses in computer science, cybersecurity, data science, and graph-based machine learning. In particular, I am highly interested in teaching undergraduate courses such as **Java Programming**, **Database Systems**, **Data Structures**, and **Algorithms**, where students develop strong foundations in computational thinking, software design, and problem-solving skills. I also enjoy teaching courses related to **Computer Networks**, **Cybersecurity Fundamentals**, **Advanced Network Security**, **Machine Learning Fundamentals**, **Graph Theory**, and **Applied Optimization**. At the graduate level, I am especially interested in developing advanced courses in **Graph Machine Learning** and **Artificial Intelligence for Security Applications**, integrating topics such as graph embeddings, graph neural networks, attack graph analytics, adversarial learning, and intelligent cyber defense systems for large-scale interconnected networks.

To help students develop strong technical foundations, I begin with clear explanations of core concepts, followed by guided examples and hands-on problem solving. Students learn best when they understand why a concept matters before they are asked to apply it. For this reason, I consistently connect theoretical material with practical motivation. In networking and cybersecurity courses, topics such as routing, VPNs, intrusion detection, and vulnerability analysis become more engaging when tied to enterprise systems, cloud platforms, and critical infrastructure. In programming courses, ideas such as modularity, recursion, and object-oriented design become clearer when students implement working systems.

Likewise, in machine learning and graph algorithms, I would connect mathematical foundations to applications such as fraud detection, recommender systems, network optimization, and automated cyber defense. My own research on attack graphs, graph learning, and security resource allocation provides many examples of how mathematics, algorithms, and AI can address modern technological challenges.

Teaching Methods

My teaching style emphasizes active learning, clarity, and engagement. Students learn most effectively when they participate rather than passively receive information. Therefore, I aim to incorporate problem-solving during lectures, guided discussions, collaborative exercises, and case studies drawn from real systems.

I also strongly believe in combining theory with implementation. Assignments and projects should reinforce classroom concepts through experimentation and design. Depending on the course, students may implement graph traversal algorithms, build Java applications using object-oriented principles, simulate network protocols, apply machine learning models to data, or analyze vulnerabilities in modeled systems.

Equally important is structured and timely feedback. Students improve most when expectations are transparent and evaluations are fair. I therefore value detailed rubrics, constructive comments, and clear guidance that helps students understand both their mistakes and the steps needed for improvement.

For advanced courses, I would also integrate research-oriented learning. Students should learn not only established methods, but also how new knowledge is created. Reading papers, presenting ideas, critiquing method-

ologies, and proposing extensions are especially valuable in fast-moving fields such as artificial intelligence.

Teaching Experience

As a Graduate Teaching Assistant at Purdue University, I have supported multiple courses across programming, engineering computing, networking, and cybersecurity. My responsibilities included holding office hours, mentoring students in laboratories, grading assignments and projects, assisting with debugging, and helping students navigate challenging technical material.

The courses I have contributed to include Communication Network Design, Advanced Network Security, Java Programming, and introductory engineering programming with Python and MATLAB. Working across these subjects has strengthened my ability to teach both theoretical and applied material while adapting to students with different backgrounds and levels of preparation.

Mentoring Philosophy

Mentoring is one of the most important responsibilities of an educator. My goal is to help students grow into independent thinkers who can approach unfamiliar problems with confidence and discipline. Effective mentoring extends beyond technical instruction; it also includes encouraging curiosity, developing communication skills, fostering professionalism, and helping students set meaningful academic and career goals.

For students interested in research, I would guide them in reading literature, identifying open problems, designing experiments, interpreting results, and presenting findings clearly. For students preparing for industry, I would emphasize software quality, teamwork, reproducibility, and scalable problem solving.

Diversity and Inclusion

As an international scholar, I deeply value inclusive learning environments where students from all backgrounds feel respected, supported, and empowered to succeed. Every student brings unique experiences and perspectives that enrich the classroom. My role as an educator is to create an atmosphere where students feel comfortable asking questions, participating in discussions, and developing confidence in their abilities.

I am especially committed to mentoring students who may lack confidence or prior exposure to technical fields. By offering guidance, patience, and high expectations, I hope to help all students recognize that success in computing and engineering is achievable through persistence, curiosity, and effort.

Future Vision

My long-term goal as an educator is to build courses and mentoring environments that combine rigorous foundations, practical relevance, and research innovation. I want students to leave my classes not only with technical knowledge, but also with the confidence to solve real problems, continue learning independently, and contribute meaningfully to society through technology.